

VIJAY WANKHEDE

Senior Smart Contract & Backend Developer

wankhede.tech@gmail.com | Blockchain · Solidity · TypeScript · NestJS · Microservices | Open to Remote

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PROFESSIONAL SUMMARY

Achievement-driven Senior Smart Contract & Backend Developer with 6+ years of experience architecting and delivering production-grade blockchain infrastructure across EVM, Bitcoin, and Solana ecosystems. Expert in Solidity smart contract development, cross-chain protocol engineering, and high-throughput TypeScript/NestJS microservices. Proven track record of measurable impact: 40% operational cost reduction, 35% gas optimization, and securing \$150M+ in delegated assets. Founding engineer at Catalysis — sole architect of core smart contract system now audit-ready for April 2026 launch. Prolific user of AI-assisted development tools (Claude with skills, Cursor) to accelerate delivery velocity while maintaining uncompromising code quality, security, and test coverage standards.

KEY ACHIEVEMENTS

 40% Operational cost reduction via API restructuring & infra optimization at Synthr	 35% Gas consumption reduction through advanced Solidity optimization techniques	 \$150M+ Delegated assets secured — Gauntlet delegation for Morpho covered vaults
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CORE TECHNICAL SKILLS

Smart Contracts & Blockchain	Solidity · EVM · Foundry · OpenZeppelin v5.4 · UUPS / ERC1538 Upgradeable Contracts · ERC20 · ERC721 · EIP-2771 · AccessControl · ReentrancyGuard · Ownable
Cross-Chain & DeFi Protocols	Axelar · LayerZero · Cross-Chain Bridges · Decentralized Oracles · Chainlink · EigenLayer · Symbiotic · Shared Security · Coverage/Insurance Protocols
Backend Engineering	TypeScript · NestJS · Node.js · REST APIs · GraphQL · gRPC · WebSockets · Microservices · Event-Driven Architecture · Kafka · High-Availability Services
Testing & Security	Foundry (Unit · Integration · Fuzz · Invariant · Mutation) · Smart Contract Audits · Vulnerability Assessments · Cryptographic Hashing · Security Best Practices
Infrastructure & DevOps	Docker · AWS · GitHub Actions · Tenderly · Defender · CI/CD Pipelines
Languages & Networks	Solidity · TypeScript · Python · Golang (scripting) · EVM Chains · Bitcoin · Solana
AI-Accelerated Development	Claude AI (with skills & advanced prompting) · Cursor · AI-assisted code review · AI-driven test generation · Rapid prototyping with LLM toolchains
Architecture & Concepts	Distributed Systems · Transaction Lifecycle Management · Concurrency Handling · Relayer/Indexer Infrastructure · Off-chain Verification · Secure Transaction Batching

PROFESSIONAL EXPERIENCE

CATALYSIS

■ **CURRENT** May 2025 – Present

Founding Engineer · Smart Contract Lead

- Sole architect & developer** of Catalysis core contract system — a modular, upgradeable smart contract suite comprising 3 managers (StakeManager, RewardsManager, SlashingManager), a secured Router middleware, and protocol adapters for EigenLayer and Symbiotic shared security protocols.
- Secured \$150M+ in delegated assets** prior to audit — obtained Gauntlet delegation for Morpho covered vaults, validating protocol trust and security posture ahead of April 2026 mainnet launch.
- Led end-to-end design, development, testing, and audit preparation of all core contracts with zero external developer dependency; brainstormed architecture directly with CTO (Dhruv Bodani).

- Implemented UUPS upgradeable proxy pattern and EIP-2771 signature-based slashing across all contracts, ensuring future-proof upgradeability and cryptographic security enforcement.
- Built comprehensive Foundry test suite covering mutation, unit, integration, fuzz, and invariant tests — achieving audit-ready code quality.
- Integrated Chainlink price feeds for multi-token vault collateral and payout pricing across heterogeneous vault structures.
- Implemented all security primitives using OpenZeppelin v5.4: AccessControl, ReentrancyGuard, Ownable, PausableUpgradeable.
- Architected coverage contracts providing institutional insurance to covered vaults (e.g., Morpho), integrating stake delegation from Gauntlet.
- **Leveraged Claude AI (with custom skills) and Cursor** to dramatically accelerate development velocity, automate boilerplate generation, and enforce high code quality standards throughout the SDLC.

SYNTHR

July 2023 – May 2025

Senior Blockchain & Backend Engineer

- **Reduced operational costs by 40%** through comprehensive API restructuring and infrastructure optimization, directly impacting protocol profitability.
- **Achieved 35% reduction in gas consumption** by applying rigorous Solidity optimization techniques including storage packing, short-circuit evaluation, and assembly-level micro-optimizations.
- Led end-to-end engineering for blockchain products: requirements, design, implementation, deployment, and developer enablement across a cross-functional team.
- Architected and delivered a production-grade cross-chain bridge leveraging Axelar and LayerZero protocols, enabling seamless multi-chain asset transfers at scale.
- Engineered pluggable ERC20/ERC721 token adapters, significantly expanding platform token compatibility and enabling rapid ecosystem integration.
- Designed and shipped developer-friendly API frameworks and SDKs providing clean abstractions for smart contract interactions and transaction lifecycle management.
- Built critical blockchain infrastructure: relayer logic, on-chain event indexing, off-chain verification pipelines, and secure transaction batching systems.
- Integrated decentralized oracle solutions for real-time pricing accuracy and reliable data feeds, enhancing protocol reliability and user trust.
- Managed complex upgradeable smart contracts using ERC1538, ensuring seamless upgrades while preserving full data integrity.
- Conducted security audits, vulnerability assessments, and conducted training sessions on cryptographic principles and security-first coding for engineering team.

INSTABIT

August 2021 – June 2023

Backend & Blockchain Developer

- Designed and architected full-stack admin and user portals for a Bitcoin-fiat exchange platform (USD/EUR ↔ BTC), serving real-time purchase and compliance workflows.
- Engineered a low-latency user experience architecture that reduced perceived wait time for 10-minute Bitcoin settlement cycles through async flow design and real-time status updates.
- Designed optimized PostgreSQL schemas for User, Purchase, KYC/AML, and Bitcoin Transaction models supporting high-write, high-concurrency workloads.
- Built all REST APIs across user, merchant, admin, and KYC/AML modules; implemented event listeners for real-time Bitcoin blockchain payment confirmation.
- Integrated hardware wallets, hot wallets, and Ledger vault services for secure BTC transfer processing and custody management.

DECODE

February 2019 – May 2021

Blockchain & Backend Developer

- Contributed to PumaPay — a crypto-native subscription billing platform — by leading database design, application architecture, and majority of API development for subscription flow and admin portal.
- Developed ERC20 smart contracts and integrated them with backend services via web3.js; built a high-throughput event listener service for token transfer processing.
- Designed PostgreSQL schemas across User, Product, Subscription, and Admin domains; implemented traffic managers for concurrent request spike handling.
- Identified and resolved a critical bug in an npm package causing irregular API responses; implemented cryptographic hashing for all passwords and sensitive data.

AI-POWERED DEVELOPMENT PHILOSOPHY

Strong advocate and practitioner of AI-assisted engineering. Regularly leverages Claude AI with custom skills for architecture review, code generation, security analysis, and documentation. Uses Cursor for intelligent IDE-level AI code completion and refactoring. Employs AI toolchains to compress development cycles, generate edge-case test scenarios, automate repetitive patterns, and maintain an uncompromisingly high bar for code quality across every phase of the SDLC — from design to deployment to audit.

NOTABLE PROTOCOLS & TECHNOLOGIES WORKED WITH

EigenLayer · Symbiotic · Axelar · LayerZero · Chainlink · Morpho · Gauntlet · PumaPay · OpenZeppelin · Foundry · Tenderly · Defender · Kafka · web3.js · NestJS · PostgreSQL · GraphQL · gRPC · WebSockets · Docker · AWS · GitHub Actions · Bitcoin Core · Solana